

ECON 370: Assignment 5

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Due: Before class on December 2nd, 2025

Names of all group members:

Directions

LLMs and AI tools may be used for this assignment, but only with careful verification of their output.

Use the Rmd template and write code to answer the following questions. In order to ensure smooth grading, I will ask you to write both your code and explanation (as comment) in code chunk. Please write your code and answer in the corresponding place.

In this homework, we will utilize simulation and test statistics to answer some theoretical questions and basic complex models.

Each question is worth three points. For each section, a “question” corresponds to a number.

Please **comment** your code clearly. 0.5 point per question will be deducted if there is a significantly lack of commenting. Also, the easier your code is to read, the more likely I will be able to figure out what you’re doing.

For any process requires randomness, you need to **set seed to 2025**. For for loops, you need to **preallocate** the result. Failing to do so will result in 0.5 point per question deduction.

Remember to **run Section O** before you proceed. After finishing all code, make sure to **restart** R session and **clear** everything from your workspace. **Rerun** everything to ensure there is no dependency issue in your code.

For **Section III**, I will let you modify some parts of the code labeled as ?. The major goal is for you to compare the results.

You may work in groups of three to four people for this homework. Please submit one script per group and clearly indicate who was in your group for the assignment.

O. Data

This is the code to extract data online and load necessary packages. Do not modify. Run this section before you proceed.

Similar to HW4, we are also creating some new data based on this formula:

$$Y = 0.5X_1 + 1X_2 + 1.5X_3 + 2X_4 + e$$

We would also have X_1 to be positively correlated to X_2, X_5 and negatively correlated to X_3 . X_1 is not correlated with X_4 at all. There are 476 variables have no contribution to Y . We are also creating an Y_2 to be created by adding random coefficients for all the 480 X variables, saved as “**X_alt**”. The actual coefficients are saved as “**b_actual**”.

I. P-Hacking for Sparse Data

We learned about p-value in *lecture 10* which can determine variable statistical significance.

To find significant results, some researchers selectively report results, run many different tests, or stopping data collection once a desired result is found. This is called **p-hacking** or **data dredging**. Let’s check if p-hacking is a good idea with simulations.

1. We have created 480 variables in “X_alt”, how many are truly related to Y ? Given that p-value is the probability that an **irrelevant** variable can be found more extreme than the given variable, with $\alpha = 0.05$, what is the chance that an **irrelevant** variable gets falsely concluded as statistically significant? Given that there are for Y , how many **irrelevant** variables would you expect to see have a low p-value?
2. Now, let’s use **for loop** to run **regressions, one variable at a time without intercept**, and save the p-values in “**Result**”. Check the proportion and number of **irrelevant** variables with $p < 0.05$. Is the proportion similar to your guess in 1?
3. If I am a researcher, who searched online and randomly (without reasoning) a list of 280 variables. I run a regression for each variable to find the useful ones. I am really “conservative” in the sense that I use $\alpha = 0.01$ and concludes that there are about **8** useful variables for Y prediction. Am I a good researcher?
4. Let’s use the idea of **training and testing** data to split the sample (I have already split them for you). Regress Y against **each 476 irrelevant variables** in the training data “X_alt_train” and “Y_train”. Find the variables significant ($p < 0.05$) and run those variables again in the testing data “X_alt_test” and “Y_test”. How many variables are still significant? Does OOS test help to flag false signals? Does OOS test eliminate “luck” (in the sense a lucky irrelevant variable is concluded as significant)?

Hint: You should skip the first 4 relevant variables when calculating the total number and proportion of significant but irrelevant variables.

5. We have seen the p-value conclusion is problematic, let’s examine OOS predictability. Take **only** the **statistically significant** variables ($p < 0.05$) from your training set, run a multiple regression in-sample (using **training set** and all “significant” variables together in **one** regression without intercept, call this “Model2”), and then use this regression to fit the test set with “**predict**” function and calculate the OOS R^2 . Try the same for just the first 4 relevant variables (call this “Model1”) and calculate the OOS R^2 again. How good are the predictions if you include irrelevant but significant variables?

Hint: For **lm** model and associated predict, you need to input a data.frame (or tibble or data.table).

6. Let’s try to use all variables in regression now and examine OOS performance. Run a multiple regression in-sample (using **training set** and all variables together in **one** regression without intercept, call

this “**Model3**”), and then use this regression to fit the test set with “**predict**” function and calculate the OOS R^2 . How good are the predictions if you just use everything without careful reasoning?

7. Let’s try the LASSO (**Model5**) and Ridge (**Model4**) with cross validation. Since this is a cross-sectional data, you can just use “**cv.glmnet**” to find the best model. (You can check lec 10 template code, remember to set seed to 2025.) Estimate the OOS R^2 for both models. Which method does better?

Hint: To fit new data with `cv.glmnet`, you should use `predict(model, newx = new_data, s = "lambda.min")`. Note for `glmnet`, you should use matrix.

8. Why do you think the Ridge and LASSO have such performance for Y ? (Hint: related to HW4 idea.)
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II. P-Hacking for Dense Data

Now, let’s repeat the exact exercise for Dense Data. Many hint can be found in the previous section.

1. We have created 480 variables in “X_alt”, how many are truly related to Y_2 ?
 2. Now, let’s use **for loop** to run **regressions, one variable at a time without intercept**, and save the p-values in “**Result_alt**”. Check the proportion and number of **significant** variables with $p < 0.05$. Why do you think there are so few significant variables?
 3. If I am a researcher, who searched online and randomly (without reasoning) a list of 480 variables. I run a regression for each variable to find the useful ones. I am really “conservative” in the sense that I use $\alpha = 0.01$ and concludes that there are about **12** useful variables for Y_2 prediction. Am I a good researcher?
 4. Let’s use the idea of **training and testing** data to split the sample (I have already split them for you). Regress Y_2 against **all 480 variables** in the training data “X_alt_train” and “Y_2_train”. Find the variables significant ($p < 0.05$) and run those variables again in the testing data “X_alt_test” and “Y_2_test”. How many variables are still significant? Does OOS test confirm their relevance? Why?
 5. Let’s try to use all variables in regression now and examine OOS performance. Run a multiple regression in-sample (using **training set** and all variables together in **one** regression without intercept, call this “**Model6**”), and then use this regression to fit the test set with “**predict**” function and calculate the OOS R^2 . How good are the predictions if you just use everything here? Guess why?
 6. Let’s try the LASSO (**Model7**) and Ridge (**Model8**) again with cross validation (“**cv.glmnet**”, remember to set seed to 2025). Estimate the OOS R^2 for both models. Which method does better?
 7. Why do you think the Ridge and LASSO have such performance for Y_2 ?
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III. Regression Trees and Neural Networks

In this section, we will try to modify **lecture 11 template code** in class for the tree models. I will provide you with analysis code about neural networks and I want you to make sense of the code and the result.

Warning: CV is much easier for cross-sectional data here. You probably need to use loop when you work

with time-series.

1. Let's just use all variables in `X_alt` to predict **Y**. Let's use **ranger** with **caret** for Random Forest + 5-fold cross validation for `Y_train`. Name this model "**Model9**". Fit the `X_alt_test` to Model9, calculate and report the OOS R^2 . How well is the Random Forest Performance?

Note: CV takes a long time.

2. Repeat with Boosting with **xgboost** to fit **Y**, use the **xgb.cv** function for cross validation with early stopping and refit the model. Please use the exact code but change to different data. Fit the `X_alt_test` to **Model10**, calculate and report the OOS R^2 . How well is the Boosting Performance?

Hint: You need to use matrix here for **xgb.cv**. Set seed twice as 2025 for CV and refitting.

Note: CV takes a long time.

3. Here is the provided Neural Networks model for **Y**. How many layers do we have? How many parameters in total do we have? How does the model work based on OOS R^2 ?
4. Let's just use all variables in `X_alt` to predict **Y_2**. Let's use **ranger** with **caret** for Random Forest + 5-fold cross validation for `Y_2_train`. Name this model "**Model12**". Fit the `X_alt_test` to Model11, calculate and report the OOS R^2 . How well is the Random Forest Performance for `Y_2`?

Note: CV takes a long time.

5. Repeat with Boosting with **xgboost** to fit **Y_2**, use the **xgb.cv** function for cross validation with early stopping and refit the model. Please use the exact code but change to different data. Fit the `X_alt_test` to **Model13**, calculate and report the OOS R^2 . How well is the Boosting Performance?

Hint: You need to use matrix here for **xgb.cv**. Set seed twice as 2025 for CV and refitting.

6. Here is the provided Neural Networks model for **Y_2**. How many layers do we have? How many parameters in total do we have? How does the model work based on OOS R^2 ?

General idea for dense data with small coefficient would be to fit a neural networks model with one layer and small number of neurons (shallow & small). Twisting an Neural Networks Model to fit data is almost an art.

7. Run the following code. I am calling it "repeated" here but that is not the official word for the idea. What single word on Lecture 11 would describe what I am doing here? How does the performance change?
8. What is your conclusion about overall OOS results for **Y** and `Y_2`? Does complicated model always work better than simple models? Why? (Hint: What is the DGP for **Y** and `Y_2`?)